

Q-Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Override DNS resolution by mapping www.spotify.com to an incorrect IP address using the local **hosts** file.

Step 1: Open the Hosts File

On Windows

1. Search for **Notepad**.
2. Right-click → **Run as administrator**.
3. Open:

C:\Windows\System32\drivers\etc\hosts

(Change the file filter from **Text Documents** to **All Files**.)

Step 2: Add an Incorrect Entry

Append the following line at the end of the file:

203.0.113.10 www.spotify.com

203.0.113.10 is a reserved documentation IP address and is intentionally not a valid Spotify server.

Save the file.

Step 3: Flush the DNS Cache (Windows)

Open **Command Prompt** as Administrator and run:

```
C:\Users\ratho>ipconfig /flushdns  
  
Windows IP Configuration  
  
Successfully flushed the DNS Resolver Cache.
```

Step 4: Ping the Domain

```
C:\Users\ratho>ping www.spotify.com

Pinging www.spotify.com [203.0.113.10] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\ratho>
```

Why This Happens

Normally:

www.spotify.com
↓
DNS Server
↓
Spotify's real IP

After editing the hosts file:

www.spotify.com
↓
Hosts File
↓
203.0.113.10 (incorrect IP)

The operating system checks the **hosts file before querying DNS**, so the incorrect mapping overrides the real DNS record.